C1.2 Cell respiration

NAME

DATE

How to use this sheet: go through the slides and videos with it, then close them and write from memory. Use your own words and add page numbers from the book (pp. 405–411) where you want more detail.

Guiding questions. What do hydrogen and oxygen contribute to releasing energy in cells? How is energy shared out and used within cells?

C1.2.1–2 ATP

What does ATP stand for? Give three properties that suit it to its role. Give one example of each use: active transport, synthesis and movement.

C1.2.3 ATP and ADP

Draw the cycle between ATP and ADP. Show where energy is released, and where it is required.

C1.2.4 Respiration and gas exchange

	CELL RESPIRATION	GAS EXCHANGE
What it is		
Where		
Purpose		

C1.2.5 Anaerobic and aerobic respiration

	ANAEROBIC	AEROBIC
Word equation		
Substrates		
Waste products		
Yield of ATP		
Where in the cell		

C1.2.6 Measuring the rate

Draw a simple respirometer and label it. Why is a CO_2 absorber needed? Which variables could you investigate?

Before you finish

Tick the ones you can explain without notes. Circle anything to ask about.

- ☐ Properties of ATP
- ☐ Uses of ATP
- ☐ ATP and ADP
- ☐ Respiration and gas exchange
- ☐ Word equations
- ☐ Anaerobic and aerobic
- ☐ Respirometer
- ☐ Rate calculations